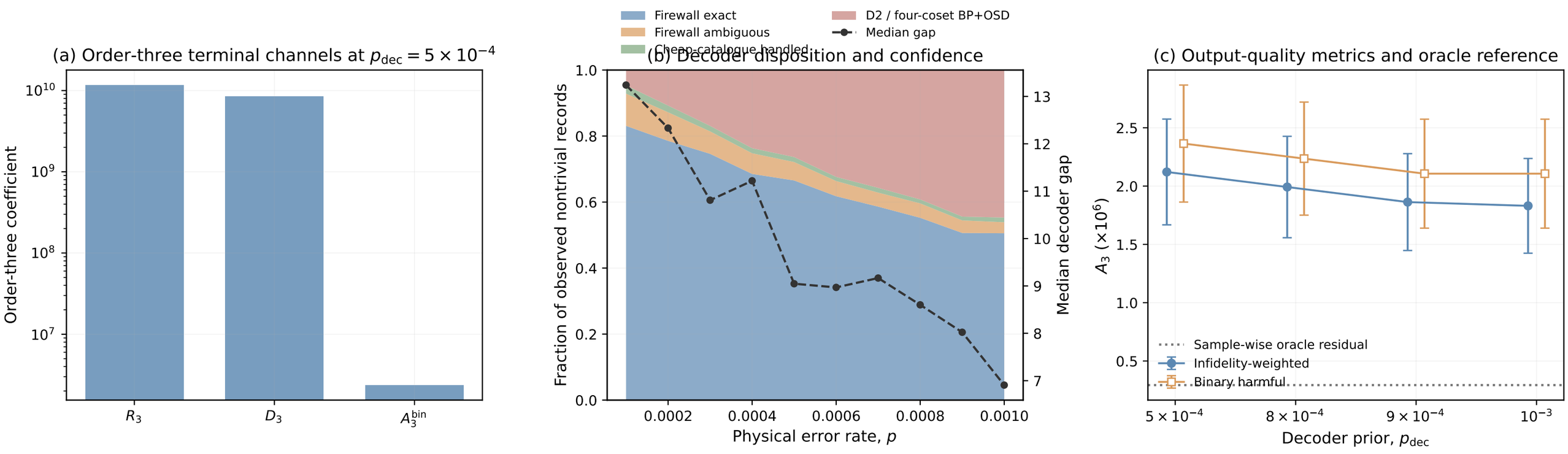
(a) Order-three terminal channels at $p_{\mathrm{dec}} = 5 \times 10^{-4}$
Order-three coefficient
10^{10}
10^{9}
10^{8}
10^{7}
R_3
D_3
A_3^{bin}
Firewall exact
Firewall ambiguous
Cheap-catalogue handled
D2 / four-coset BP+OSD
Median gap
(b) Decoder disposition and confidence
Fraction of observed nontrivial records
Median decoder gap
Physical error rate, p
0.0002
0.0004
0.0006
0.0008
0.0010
(c) Output-quality metrics and oracle reference
A_3 ($\times 10^6$)
Sample-wise oracle residual
Infidelity-weighted
Binary harmful
5×10^{-4}
8×10^{-4}
9×10^{-4}
10^{-3}
Decoder prior, p_{dec}